

Monel R405 | High Strength & Machinable Nickel-Copper Alloy

Introduction:

Monel R405 (UNS N04405) is a high-strength nickel-copper alloy with excellent corrosion resistance and superior machinability due to controlled sulfur content. Ideal for precision components in marine, chemical, and industrial applications, R405 offers toughness across a wide temperature range.

Chemical Composition:

Element	Composition (%)
Nickel	63.0 min
Copper	28.0 - 34.0
Carbon	0.3 max
Manganese	2.0 max
Iron	2.5 max
Sulfur	0.025 - 0.060
Silicon	0.5 max

Physical & Mechanical Properties:

- **Density:** 8.80 g/cm³
- **Melting Range:** 1300-1350 °C
- **Yield Strength:** 25-105 ksi (depending on treatment)
- **Tensile Strength:** 35-50 ksi
- **Elongation:** 70-115%
- **Hardness:** 110-245 HB

Available Products:

- Seamless & Welded Pipes (ERW & EFW)
- Tubes (Capillary, U-Bend)
- Sheets, Plates, Coils & Strips
- Round, Flat, Hex & Square Bars
- Flanges, Fittings, Fasteners
- Custom Machined Components

Applications:

- Automatic Screw Machine Parts (bolts, shafts, connectors)
- Chemical & Petrochemical Equipment (valves, pumps, heat exchangers)
- Marine & Offshore Components (shaft sleeves, seawater valves)
- Oil & Gas Industry (drill collars, downhole tools)
- Nuclear & Power Systems (steam turbines, auxiliary components)
- Defense & Aerospace Precision Fittings
- Food & Pharmaceutical Equipment (valves, connectors)

Advantages:

- Excellent machinability & corrosion resistance
- High strength at cryogenic & elevated temperatures
- Cost-effective for precision components
- Resistant to seawater, acids, alkalis, and high-temperature steam

Why Choose Moksh Tubes & Fittings LLP:

- ISO 9001:2015 certified
- Specialized stockist & global supplier of Monel R405
- Custom machining support & quick delivery
- Full traceability with Mill Test Reports